# Odds & Ends: Multiple Linear Regression

Categorical Predictors + Adjusting for Confounding

Emile Latour

March 18, 2026

# From SLR to Multiple Regression

# Where we left off

In the SLR lecture we asked:

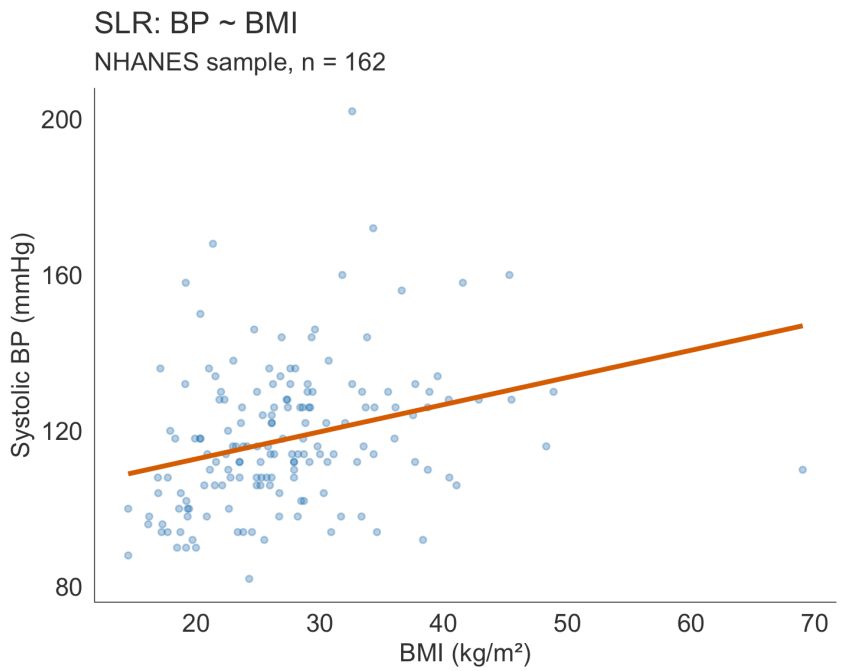
> **Does BMI predict systolic blood pressure?**

We fit:

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 \cdot \text{BMI}$$

And found: for every 1 kg/m² increase in BMI, systolic BP is expected to increase by 0.70 mmHg on average.

But real biology is messier. **Multiple variables** are usually at play — age, sex, medications, comorbidities...



# Roadmap for Today

**Part 1: Categorical Predictors**

- A new kind of explanatory variable
- The binary categorical model
- Connecting to what you already know
- Interpreting output in R
- Changing the reference category

**Part 2: Multiple Linear Regression**

- Adding predictors to the model
- Confounding: why adjustment matters
- Fitting MLR in R with `lm()`
- Interpreting adjusted coefficients
- $R^2$ and adjusted $R^2$
- Unadjusted vs. adjusted: a side-by-side comparison

# Part 1: Categorical Predictors

# A new question about blood pressure

So far, the explanatory variable has always been **continuous** (e.g., BMI, Age).

What if the explanatory variable is **categorical**?

**New question**

**Does mean systolic blood pressure differ between males and females in the NHANES sample?**

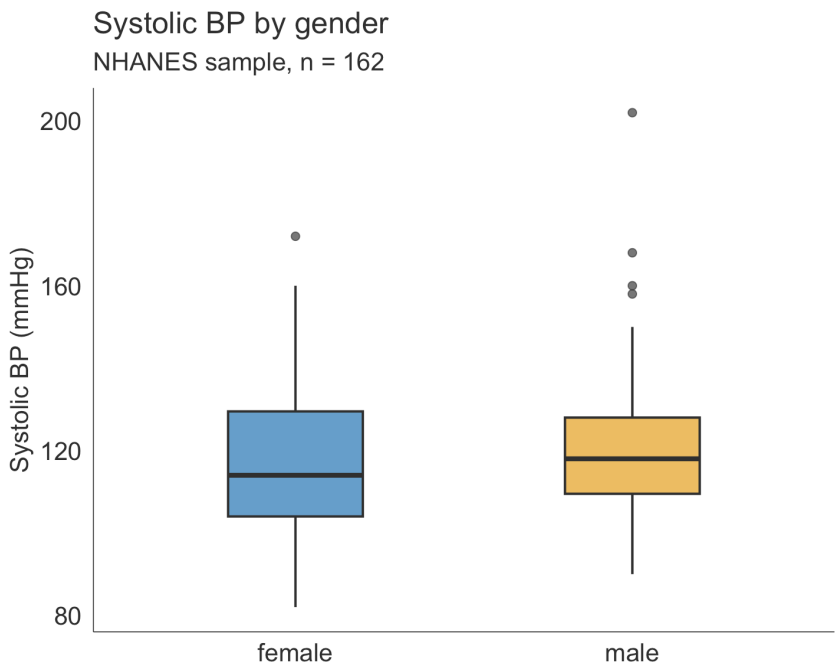
- Response $Y$: Systolic BP (continuous)
- Explanatory $X$: Gender (categorical: female / male)

You might think: *"I know how to do this — it's a two-sample t-test!"* And you're right. But regression gives us a unified framework that extends naturally to multiple predictors.

# Visualizing BP by gender

| Gender | n | Mean BP | SD |
|---|---|---|---|
| female | 78 | 116.3 | 18.4 |
| male | 84 | 119.3 | 18.0 |

Males have a slightly higher mean BP than females in this sample.



# The regression model with a binary categorical predictor

For a binary categorical variable, R creates an **indicator variable** (0/1) automatically:

$$\text{Gender}_{\text{male}} = \begin{cases} 1 & \text{if male} \\ 0 & \text{if female} \end{cases}$$

The model becomes:

$$E(\text{BP}) = \beta_0 + \beta_1 \cdot \text{Gender}_{\text{male}}$$

**Intercept $\hat{\beta}_0$**

The **estimated mean BP for females** — the group where $\text{Gender}_{\text{male}} = 0$.

This is the **reference category**.

**Slope $\hat{\beta}_1$**

The **estimated difference in mean BP** comparing males to females.

$\hat{\beta}_1 = \overline{Y}_{\text{male}} - \overline{Y}_{\text{female}}$

# Running the model in R

```r
model_cat <- lm(BPSys1 ~ Gender, data = nhanes_bp)

broom::tidy(model_cat, conf.int = TRUE)
```

```
# A tibble: 2 × 7
  term        estimate std.error statistic   p.value conf.low conf.high
  <chr>          <dbl>     <dbl>     <dbl>     <dbl>    <dbl>     <dbl>
1 (Intercept)   116.        2.06     56.4  1.61e-107   112.      120.
2 Gendermale      3.00      2.86      1.05 2.96e-  1    -2.65      8.66
```

**Interpretation**

- **Intercept** (116.3 mmHg): The estimated mean systolic BP for **females**.
- **Gendermale** (3.0 mmHg): Males have, on average, a systolic BP that is 3.0 mmHg **higher** than females (95% CI: -2.7 to 8.7 mmHg).

The p-value (0.296) indicates this difference is not statistically significant.

# This is a t-test in disguise

**Connecting to what you already know**

Running `lm(BPSys1 ~ Gender)` with a binary predictor gives you the **same p-value** as a two-sample t-test (assuming equal variances).

Regression is the more general framework — it lets us add more predictors, which a t-test cannot do.

```r
# Two-sample t-test for comparison
t.test(BPSys1 ~ Gender, data = nhanes_bp, var.equal = TRUE)
```

```
	Two Sample t-test

data:  BPSys1 by Gender
t = -1.0485, df = 160, p-value = 0.296
alternative hypothesis: true difference in means between group female and group male is not equal to 0
95 percent confidence interval:
 -8.655674  2.652011
sample estimates:
mean in group female   mean in group male 
            116.3077             119.3095 
```

# Changing the reference category

R chooses the reference category **alphabetically** — "female" comes before "male", so female is the reference.

To change it, use `relevel()`:

```r
nhanes_bp_male_ref <- nhanes_bp %>%
  mutate(Gender = fct_relevel(Gender, "male"))  # base R equivalent: relevel(Gender, ref = "male")

lm(BPSys1 ~ Gender, data = nhanes_bp_male_ref) %>%
  tidy(conf.int = TRUE)
```

```
# A tibble: 2 × 7
  term         estimate std.error statistic   p.value conf.low conf.high
  <chr>           <dbl>     <dbl>     <dbl>     <dbl>    <dbl>     <dbl>
1 (Intercept)    119.        1.99     60.1  1.14e-111   115.      123.
2 Genderfemale    -3.00      2.86     -1.05 2.96e-  1    -8.66      2.65
```

**Note**

The **estimate changes sign** but the **magnitude and p-value are identical** — we're just asking the question from the other direction. Use whichever reference group makes the most scientific sense.

# Part 2: Multiple Linear Regression

# Adding more predictors

We have two questions we'd like to answer simultaneously:

- Is BMI associated with BP?
- Is gender associated with BP?

**And**: we know BMI and gender are themselves related — males tend to have different BMI distributions than females. This creates a problem called **confounding**.

The solution is to put both predictors in the same model:

$$E(\text{BP}) = \beta_0 + \beta_1 \cdot \text{BMI} + \beta_2 \cdot \text{Gender}_{\text{male}}$$

**Multiple linear regression (MLR)**

A regression model with **two or more** explanatory variables. Each coefficient describes the association between that predictor and the response, **holding the other predictors constant**.

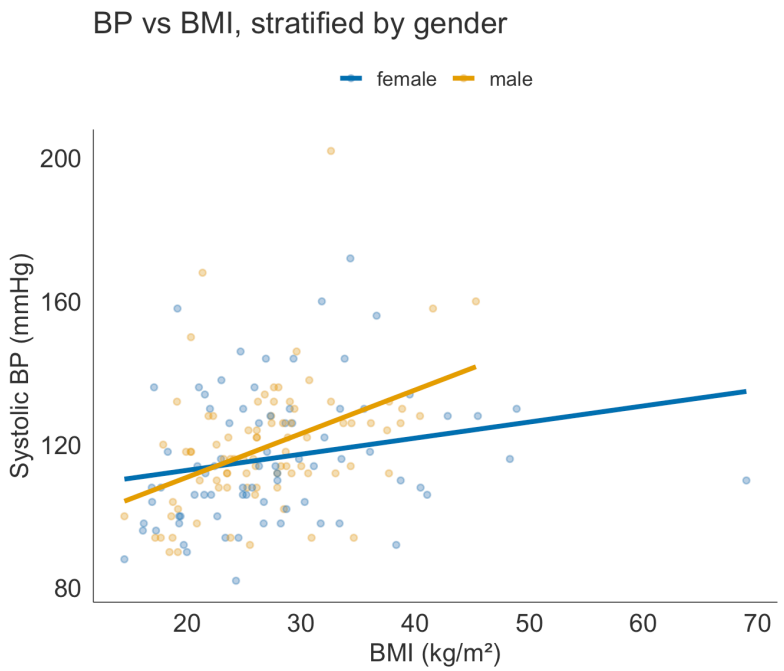
# What is confounding?

A **confounder** is a variable that is:

1. Associated with the **explanatory variable** of interest
2. Associated with the **response variable**
3. Not on the causal pathway between them

**Example here:** Gender is associated with both BMI and BP.

If we only look at BP ~ BMI, some of the apparent "BMI effect" is really a "gender effect" in disguise.




### Adjusting for confounding

Including the confounder in the model lets us estimate the association between BMI and BP **among people of the same gender** — a fairer comparison.

# Fitting MLR in R

The syntax extends naturally from SLR — just add predictors with `+`:

```r
model_mlr <- lm(BPSys1 ~ BMI + Gender, data = nhanes_bp)

tidy(model_mlr, conf.int = TRUE)
```

```
# A tibble: 3 × 7
  term        estimate std.error statistic  p.value conf.low conf.high
  <chr>          <dbl>     <dbl>     <dbl>    <dbl>    <dbl>     <dbl>
1 (Intercept)   96.6       5.32      18.2  1.16e-40   86.1      107.
2 BMI            0.710     0.178      3.99 1.02e- 4    0.358      1.06
3 Gendermale     3.58      2.74       1.31 1.93e- 1   -1.83       9.00
```

**Note**

R handles the categorical variable automatically. You do not need to create indicator variables by hand — just make sure `Gender` is a `factor`.

# Interpreting the adjusted coefficients

$$\widehat{\text{BP}} = \hat{\beta}_0 + \hat{\beta}_1 \cdot \text{BMI} + \hat{\beta}_2 \cdot \text{Gender}_{\text{male}}$$

```r
tidy(model_mlr, conf.int = TRUE)
```

```
# A tibble: 3 × 7
  term        estimate std.error statistic  p.value conf.low conf.high
  <chr>          <dbl>     <dbl>     <dbl>    <dbl>    <dbl>     <dbl>
1 (Intercept)   96.6       5.32      18.2  1.16e-40   86.1      107.
2 BMI            0.710     0.178      3.99 1.02e- 4    0.358      1.06
3 Gendermale     3.58      2.74       1.31 1.93e- 1   -1.83       9.00
```

**Key rule: always say "holding the other variables constant"**

- **BMI** (0.71 mmHg): For every 1 kg/m² increase in BMI, systolic BP is expected to increase by 0.71 mmHg on average, **holding gender constant**.
- **Gendermale** (3.58 mmHg): Males have, on average, a systolic BP that is 3.58 mmHg higher than females, **holding BMI constant** (i.e., comparing males and females at the same BMI).
- **Intercept**: Expected mean BP for a **female** with BMI = 0. (Not a meaningful prediction — extrapolation.)

# Unadjusted vs. adjusted: what changed?

### Unadjusted (SLR): BP ~ BMI

| Term | Est. | Lower | Upper | p-value |
|---|---|---|---|---|
| (Intercept) | 98.83 | 88.85 | 108.81 | < 0.001 |
| BMI | 0.70 | 0.35 | 1.05 | < 0.001 |

BMI slope: **0.70** mmHg per kg/m²

### Adjusted (MLR): BP ~ BMI + Gender

| Term | Est. | Lower | Upper | p-value |
|---|---|---|---|---|
| (Intercept) | 96.63 | 86.13 | 107.13 | < 0.001 |
| BMI | 0.71 | 0.36 | 1.06 | < 0.001 |
| Gendermale | 3.58 | -1.83 | 9.00 | 0.193 |

BMI slope: **0.71** mmHg per kg/m²

### Tip

The BMI coefficient changed slightly after adjusting for gender — a sign that gender was acting as a mild confounder. In some datasets, adjustment can dramatically change an estimate (or even flip its direction).

# $R^2$ and adjusted $R^2$

You've seen $R^2$ as a measure of model fit. In MLR there's a wrinkle:

**$R^2$ always increases when you add predictors**

Even if a new predictor is pure noise, $R^2$ can only stay the same or go up — making it a poor tool for comparing models with different numbers of predictors.

**Adjusted** $R^2$ penalizes for the number of predictors:

$$R^2_{\text{adj}} = 1 - \frac{(1 - R^2)(n - 1)}{n - p - 1}$$

where $p$ = number of predictors. A new predictor only improves adjusted $R^2$ if it adds more than it costs.

```r
glance(model_mlr) %>%
  select(r.squared,
         adj.r.squared,
         sigma,
         nobs)
```

```
# A tibble: 1 × 4
  r.squared adj.r.squared sigma  nobs
      <dbl>         <dbl> <dbl> <int>
1    0.0970        0.0857  17.4   162
```

In our model: $R^2$ = 0.097, adjusted $R^2$ = 0.086.

# The MLR workflow

### 1. Visualize

```
ggplot(data, aes(x = X1, y = Y, color
  geom_point() +
  geom_smooth(method = "lm")
```

### 2. Ensure categorical vars are factors

```
data <- data %>%
  mutate(X2 = factor(X2))
```

### 3. Fit

```
model <- lm(Y ~ X1 + X2, data = data)
```

### 4. Coefficients + inference

```
tidy(model, conf.int = TRUE)
```

### 5. Model fit

```
glance(model)   # R², adj. R², sigma
```

### 6. Change reference category (if needed)

```
data <- data %>%
  mutate(X2 = foracts::fct_relevel(X2, "your_ref"))
```

### 7. Interpret everything as "adjusted"

Every coefficient describes the association with $Y$ **holding all other predictors constant**.

# Categorical predictors with 3+ levels

What if a categorical variable has **more than two levels**?

R creates **one coefficient per level, each compared to the same reference group** — the reference category is absorbed into the intercept and doesn't get its own row.

Let's create age groups from the continuous `Age` variable and fit the model (next slide):

```r
nhanes_bp <- nhanes_bp %>%
  mutate(Age_group = cut(Age,
                         breaks = c(0, 39, 59, Inf),
                         labels = c("Under 40", "40–59", "60+")))

nhanes_bp %>%
  janitor::tabyl(Age_group)
```

```
Age_group  n   percent
 Under 40 95 0.5864198
    40–59 43 0.2654321
      60+ 24 0.1481481
```

# Interpreting the output

```r
model_age <- lm(BPSys1 ~ Age_group, data = nhanes_bp)

tidy(model_age, conf.int = TRUE)
```

```
# A tibble: 3 × 7
  term           estimate std.error statistic   p.value conf.low conf.high
  <chr>             <dbl>     <dbl>     <dbl>     <dbl>    <dbl>     <dbl>
1 (Intercept)       110.       1.62     68.1  1.60e-119    107.      113.
2 Age_group40-59     16.2      2.90      5.60 9.39e-  8     10.5      22.0
3 Age_group60+       22.1      3.61      6.14 6.36e-  9     15.0      29.3
```

### Reference category: "Under 40"

The intercept (110.3 mmHg) is the estimated mean BP for participants **under 40** — the group that doesn't get its own row.

### Interpretation template

- **40–59**: mean BP is 16.2 mmHg **higher than the under-40 group**
- **60+**: mean BP is 22.1 mmHg **higher than the under-40 group**

Every coefficient is *always* relative to the reference — never relative to each other.

# What we haven't covered (and where to go)

MLR is a huge topic. Today's lecture gives you the foundation; there's much more in the textbook (Chapter 7) and beyond:

### In your textbook (Ch. 7)

- Model selection strategies (§7.8)
- Interaction terms (§7.7)
- Connection to ANOVA (§7.9)

### Where regression leads

- **Logistic regression** — binary response (yes/no, alive/dead)
- **Survival analysis** — time-to-event outcomes
- **Mixed models** — correlated observations